QR code decoder (mkQRDecoder): physical design report

CS6230 CAD for VLSI, 2026. Every number below is copied from Design Compiler or IC Compiler output; the run folders are listed in section 7.

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1.00 ns	71.90%	19,724 um^2	14.71 mW
f2f slack 0.00 (post-place)	utilization (core 170 x 170 um)	post-place cell area	post-place power

1. Setup

- Design: mkQRDecoder at the delivered buffer size (CAP = 1024 bytes). Each of the four designs differs from the previous one in exactly one BSV file (section 3).
- Synthesis: Design Compiler U-2022.12, non-topographical, library saed14rvt_base_ss0p72v125c (unchanged from the udcounter example), wire-load model (35000), 6 cores. Clock CLK; input and output delay on every port except the clock = 40% of the period; path groups f2f, i2f, f2o, i2o; flow derived from the udcounter scripts, with compile_ultra -no_autoungroup -no_seq_output_inversion -timing_high_effort_script.
- Placement: IC Compiler (V-2023.12), placement only (place_opt, no clock-tree synthesis, no routing), **170 x 170 um core**, libraries and power-strap layers as in the udcounter flow. The core size was chosen so that the post-PlaceOpt utilization is at least 70% (the first run used a 210 x 210 um core and gave 47.7%; the netlist shrinks during place_opt, so the core was sized down to the placed cell area).
- Definitions: critical-path period and slack are from the f2f group of report_qor. Max clock frequency is the smallest period for which our runs met timing (slack 0 or better, zero violating paths). Power is from report_power with default switching activity, so it compares designs but is not an absolute figure. "Design area" in Table 5 is DC's total, which includes a wire-load estimate of net area; "cell area" is cells only and is what the placement tool reports.

2. Table 5: post-synthesis QoR

Design	Change vs previous	Clk period (ns)	f2f slack (ns)	Max freq (MHz)	Leaf cells	Seq. cells	Power (mW)	Design area	Cell area
1	baseline	2.7	0.00	370.4	42,275	9,496	6.145	64,013.0	21,310.8
2	format decode (QRFormat)	2.1	0.00	476.2	40,862	9,496	7.780	62,688.1	20,381.8
3	GF inverse as table (QRGF)	1.0	0.00	1000	41,830	9,496	15.156	64,196.2	26,896.4
4	registered buffer address (QRRS)	1.0	0.00	1000	40,408	9,508	15.653	61,136.6	24,828.2

Each row is the smallest period at which that design met timing among the runs we made. The rows are at different periods, so power and area are not comparable down the table; section 4 compares designs at the same period. Design 1 missed 2.5 ns by 0.11 ns (f2f path 2.59 ns), so 2.7 ns is its first passing run. Design 2 missed 1.9 ns by 0.013 ns, so its true limit lies between 1.9 and 2.1 ns. Designs 3 and 4 both missed 0.95 ns (section 4).

3. The four iterations

Each iteration followed the same loop: read the timing report, find what limits the clock, change one thing, check in simulation that behaviour is unchanged, synthesize again.

#	What the previous reports showed	Change (one file)	Result
1	Baseline. At 2.5 ns the five worst f2f paths all start at the second format copy (rd/r_fmtW2_reg_7_) and end at maskR/lvlR (-0.11 ns, 58 violating paths). The format decode compared both copies against all 32 valid words with a 32-deep compare-and-select chain.	(none)	Meets 2.7 ns.
2	As above.	QRFormat.bsv: replace the chain with 32 independent "within 3 bits" tests. Valid words differ in at least 7 bits, so at most one can be within 3 and the nearest-word search is an OR of the match. Checked exhaustively in Python and by the format unit test (all 2^15 words, 400,000 random pairs).	Meets 2.1 ns. Misses 1.9 ns by 0.013 ns (43 violating paths).
3	At 1.9 ns the five worst paths run from rs_dReg to rs_invv/rs_binv: the GF(2^8) inverse, built as a^254 from six multipliers in series (about 40 XOR levels).	QRGF.bsv: gfInv becomes a 256-entry table built at compile time from the old chain (kept as gfInvChain), so the values are the same by construction. The GF unit test compares all 256 inverses with the model.	Meets 1.0 ns. At 1.9 ns: cell area -5.9%, leaf cells -2.5% against design 2.
4	At 1.0 ns three of the five worst f2f endpoints are bb_mem_rdReg (start points rs_mR, rs_reqIdx): a subtract and an add sit in front of the 1000-way read decode.	QRRS.bsv: a 12-bit register rdAddr holds the buffer address, loaded one state earlier (Chien pass) or incremented with reqIdx (syndrome and output passes). No state is added, so cycle counts are identical.	Meets 1.0 ns. At 1.0 ns: cell area -7.7%, leaf cells -3.4%, power +3.3% against design 3. At 0.95 ns it misses by 0.07 ns, where design 3 misses by 0.49 ns.

Behaviour check for every iteration: the whole-decoder regression (1,614 symbols, 1,168 decode and 446 must fail, 1024-byte buffer) gives 0 mismatches, and the latency is identical for all four designs (max 37,388 cycles, mean 11,278), since none of the changes alters the number of cycles. For designs 3 and 4 the Verilog that was synthesized was checked to be identical (apart from the compiler timestamp line) to the Verilog generated by the tested build.

4. Designs compared at the same period

Period	Design	f2f slack (ns)	Violating paths	Leaf cells	Power (mW)	Design area	Cell area
2.1 ns	2	0.00	0	40,862	7.780	62,688.1	20,381.8
2.1 ns	3	0.00	0	39,940	7.357	61,908.9	20,369.6
1.9 ns	2	-0.01 (-0.013)	43	40,702	8.333	63,360.3	21,141.9
1.9 ns	3	0.00	0	39,695	8.200	61,793.8	19,900.4
1.0 ns	3	0.00	0	41,830	15.156	64,196.2	26,896.4
1.0 ns	4	0.00	0	40,408	15.653	61,136.6	24,828.2
0.95 ns	3	-0.49	9,506	54,109	17.124	69,517.3	26,888.6
0.95 ns	4	-0.07	2,440	40,471	16.052	62,279.7	26,193.0
0.90 ns	3	-0.45	9,507	48,881	17.470	65,964.3	26,745.6
0.90 ns	4	-0.03	339	41,350	16.944	64,192.8	27,184.2

Reading the table. Design 3 against design 2: smaller at every period and it closes timing where design 2 does not. Design 4 against design 3 at 1.0 ns: smaller cells and area, slightly more power, same frequency; below 1.0 ns neither closes timing, but design 4 misses by far less. Results of runs that miss timing are noisy, because the tool stops in a different state each time, and the results are not smooth in the period: design 4 at 1.1 ns came out with 51,880 leaf cells and 68,747.1 total area (cell area 25,962.0, power 13.885 mW), more than at 1.0 ns. We therefore compare at the periods shown and do not read trends from single runs.

5. Table 6: post-placement QoR (best design: design 4 at 1.0 ns)

Design	Clk period (ns)	f2f slack (ns)	Max freq (MHz)	Leaf cells	Seq. cells	Power (mW)	Block dimensions (um)	Design area (cell area)	Utilization
4 (best)	1.00	0.00	1000	38,074	9,508	14.710	169.98 x 169.23	19,724.2	71.90%

	Synthesis (DC)	Placement (ICC)
f2f critical path length	0.97 ns	0.98 ns
Leaf cells / sequential cells	40,408 / 9,508	38,074 / 9,508
Total power	15.653 mW	14.710 mW (dynamic 14.669, leakage 0.041)
Cell area	24,828.2	19,724.2
Std-cell utilization	n/a	71.90% (444,253 of 617,893 sites)
Other groups all met		f2o slack +0.05 ns, i2f 0.00 ns; no i2o group in the report

Timing is still met at 1.0 ns after placement, so there is no degradation in frequency; power and cell area are lower than after synthesis because place_opt sizes cells against real placement while synthesis used a wire-load model. The core is 170 x 170 um (269 rows) and the post-PlaceOpt core utilization is 71.90% (38,074 cells), which meets the 70% minimum. The legality check reports 0 cells off row, 0 overlaps, 0 blockage overlaps and 0 orientation, site or power-strap violations.

6. Limits of these numbers

- Post-placement figures use an ideal clock (no clock-tree synthesis) and wire parasitics estimated from the placement, since there is no routing. Skew and routed-wire delay would reduce the margin.
- Synthesis uses a wire-load model, so absolute area and timing differ from a physical flow; ICC's cell area is not comparable with DC's "design area", which includes estimated net area.
- Power uses default switching activity. It is meaningful between the designs, not as an absolute figure.
- The SDC files in pd/ are the constraints written by each DC run with the expanded group_path register lists removed (they list every register by synthesis name and make the file about 1.5 MB). The clock and the 13 input and 32 output delays are unchanged, and the path groups are defined in each design_N_dc.tcl. best_design_place.sdc is the same constraint set that ICC read.
- The ICC log (best_icc_run.log, about 48 MiB) is submitted unfiltered. Most of it is repeated power-routing warnings (PGRT, "wire dropped") from the power straps and rails; the legality report is clean and the log contains no errors.
- The "DRC and LVS clean" wording of the project document cannot be checked after a placement-only flow; the legality check above is the closest check available.

7. Which run is which

Submitted as	BSV source folder	DC run tag	Server folder
design_1	modules (baseline)	design_1_p27	properrun1
design_2	run2_modules (+ QRFormat)	design_1_p21	QR_code_accellerator_run2
design_3	run3_modules (+ QRGF)	design_1_p10	QR_code_accellerator_run3
design_4, best_design	run4_modules (+ QRRS)	design_4_p10	QR_code_accellerator_run4 (placement in icc1/)

The tags kept their early names, so the tag does not always match the design number. The BSV in run2_modules differs from modules in one file (QRFormat.bsv), run3_modules from run2_modules in one file (QRGF.bsv), and run4_modules from run3_modules in one file (QRRS.bsv).